Quadratic Kernel Mapping and Kernel Trick

$a = [a_1, a_2]^T \quad b = [b_1, b_2]^T$

$(a^T b)^2 = (a_1 b_1 + a_2 b_2)^2 = a_1^2 b_1^2 + 2a_1 a_2 b_1 b_2 + a_2^2 b_2^2$

□ **$(a^T b)^2 = a_1^2 b_1^2 + 2a_1 a_2 b_1 b_2 + a_2^2 b_2^2$ (boxed per vision)**

$\varphi(a)^T \varphi(b) = [a_1^2, \sqrt{2}\, a_1 a_2, a_2^2] \cdot [b_1^2, \sqrt{2}\, b_1 b_2, b_2^2]$ → **$\varphi(a)^T \varphi$**

$= a_1^2 b_1^2 + 2a_1 a_2 b_1 b_2 + a_2^2 b_2^2$

$\varphi(x) = [x_1^2, \sqrt{2}\, x_1 x_2, x_2^2]^T$ → **$\varphi(x)$ d**

2D → 3D

$a = [1, 2]^T \quad b = [3, 4]^T$

$(a^T b)^2 = (1 \cdot 3 + 2 \cdot 4)^2 = (3 + 8)^2 = 11^2 = 121$

$\varphi(a)^T \varphi(b) = 1^2 \cdot 3^2 + 2 \cdot 1 \cdot 2 \cdot 3 \cdot 4 + 2^2 \cdot 4^2 = 9 + 48 + 64 = 121$

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$[a^T b]^2 = (1 \cdot 3 + 2 \cdot 4)^2 = 11^2 = 121$